Product Cost-Quality Scoring Framework

A practical method to rank products so that low cost and high quality receive the best scores, while products below specification - especially expensive ones - receive the lowest scores.

Recommended design
Combine a normalized cost score with a capped quality score, then apply a strong penalty whenever quality falls below the minimum specification.

1. Variables

Symbol	Meaning
c_i	Cost of product i
q_i	Quality of product i
q_{min}	Minimum acceptable quality specification
q_{target}	Quality level above which extra quality receives no additional score
c_{min}, c_{max}	Reference minimum and maximum cost values

2. Cost score

Normalize cost so that the cheapest product receives 1 and the most expensive receives 0:

$$CostScore_i = (c_{max} - c_i) / (c_{max} - c_{min})$$

Lower cost therefore always improves the score.

3. Quality score

Reward quality above specification, but cap the benefit at a meaningful target. This prevents excessive quality or overprocessing from being rewarded indefinitely.

$$QualityScore_i = clip[(q_i - q_{min}) / (q_{target} - q_{min}), 0, 1]$$

Quality position	Quality score
Below q_{min}	0 before applying the out-of-specification penalty
At q_{min}	0
Between q_{min} and q_{target}	Increases linearly from 0 to 1
At or above q_{target}	1

4. Out-of-specification penalty

Meeting specification is normally non-negotiable. A separate penalty ensures that a cheap but non-compliant product does not outrank a compliant product.

$$Penalty_i = 0, \text{ if } q_i >= q_{min}$$

$$Penalty_i = P_0 + P_1(q_{min} - q_i) / q_{min}, \text{ if } q_i < q_{min}$$

P_0 is a fixed penalty for any specification failure. P_1 increases the penalty according to the severity of the quality shortfall.

5. Final recommended score

Using equal weights as a starting point:

$$Score_i = 50 \times CostScore_i + 50 \times QualityScore_i - Penalty_i$$

The weights can be adjusted. For example, use 60% quality and 40% cost when quality should dominate, or 60% cost and 40% quality when all evaluated products are already reliably compliant.

6. Expected ranking behaviour

Product profile	Expected score
High quality, low cost	Highest
Acceptable quality, low cost	High
High quality, high cost	Intermediate
Below-specification quality, low cost	Low
Below-specification quality, high cost	Lowest

7. Suggested starting parameters

Parameter	Starting recommendation
Cost weight	50
Quality weight	50
q_{target}	A meaningful quality target above specification, not the observed maximum
P_0	50 points, making any specification breach immediately visible
P_1	50 points, distinguishing minor from severe quality failures
Cost reference range	Prefer robust limits such as the 5th and 95th percentiles rather than raw extremes

8. Important implementation note

Use stable reference values for normalization. If c_{min} and c_{max} are recalculated from every small batch, the same product may receive a different score depending on the comparison set. For operational use, define the cost limits from a sufficiently long historical period and review them periodically.

Interpretation: The resulting score is a ranking index, not a physical measurement. Its parameters should be validated against business priorities and known examples of good and bad products.